CIA World Factbook Archive

A Comprehensive Digital Preservation and Intelligence Analytics Platform

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Abstract

This report documents the design, construction, and deployment of the CIA World Factbook Archive — a research platform that preserves, structures, and visualizes 36 years of the Central Intelligence Agency's *World Factbook* (1990–2025). The project ingests raw source material spanning six distinct text formats, five HTML layouts, and modern JSON feeds, normalizes over one million data points across 281 geopolitical entities into a unified SQLite database, and exposes the result through a full-featured web application with interactive analytics, geospatial visualization, and a custom dashboard builder. The archive represents the only publicly available system that makes every edition of the CIA World Factbook queryable, comparable, and downloadable in structured form.

Key Metrics: 281 Entities | 36 Editions | 9,536 Country-Year Records | 1,071,603 Data Points

Live URL: https://worldfactbookarchive.org

Source Code: https://github.com/MilkMp/CIA-World-Factbooks-Archive-1990-2025

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Chapter 1

Genesis and Vision

1.1 Background

The CIA *World Factbook* is the United States Intelligence Community's flagship open-source reference on every country and territory in the world. Published annually since 1962 and made freely available to the public since 1971, it has served as the de facto standard reference for geopolitical, demographic, economic, and military data used by government analysts, journalists, researchers, and educators worldwide.

Despite its importance, no centralized archive existed that preserved every edition in structured, machine-readable form. Older editions published as plain text files in the 1990s were scattered across university FTP mirrors and the Internet Archive. HTML editions from the 2000s and 2010s were accessible only through the Wayback Machine, often with broken links and inconsistent formatting. The CIA itself maintained only the current edition on its website, with each new release overwriting the previous one.

On **February 4, 2026**, the CIA discontinued publication of the World Factbook entirely, making the preservation of historical editions an urgent priority.

1.2 The Problem

Without a structured archive, researchers faced several critical barriers:

1. **No longitudinal analysis.** Comparing a country's GDP trajectory or military spending over two decades required manually downloading and parsing dozens of files in incompatible formats.
2. **Format fragmentation.** The Factbook changed its publication format at least 12 times between 1990 and 2025, from pipe-delimited text to nested HTML to structured JSON.
3. **Entity instability.** Countries dissolved (Soviet Union, Yugoslavia), merged (Germany), renamed (Swaziland to Eswatini), and changed codes across editions.
4. **Field vocabulary drift.** The CIA renamed, merged, and restructured hundreds of data fields over the decades — "Total area" became "Area," "National product" became "Real GDP (purchasing power parity)," and so on.
5. **Disappearing sources.** University mirrors went offline, the Wayback Machine returned incomplete captures, and the CIA's own site offered no historical access.

1.3 Vision

The CIA World Factbook Archive was conceived as a comprehensive digital preservation and analytics platform with three objectives:

1. **Preserve** every edition of the World Factbook from 1990 through its final 2025 publication in a single, structured, queryable database.
2. **Normalize** the data across format changes, field renames, and entity reorganizations so that any indicator can be compared across any country and any year.
3. **Visualize** the data through an interactive web application that makes 36 years of intelligence-grade geopolitical data accessible to researchers, students, journalists, and the general public.

1.4 Development Timeline

The project was built in an intensive 14-day development sprint, producing 205 commits across two repositories. The following timeline is derived from the project's git history:

Date	Milestone
Feb 16, 2026	Initial release: ETL pipeline, SQLite database, GitHub Pages static site
Feb 16	Full web application committed (FastAPI + Jinja2 + Mapbox GL JS)
Feb 18–19	Migration from Plotly.js to Mapbox GL JS; v2 data drop (1996 repairs, analytics explorer)
Feb 20	MIT license adopted; visitor analytics system added
Feb 22	Webapp split to private repository; terrain/fog rendering; custom domain setup
Feb 23	7-layer security system; header navigation redesign; Intelligence Atlas; political dashboard; worldfactbookarchive.org launched
Feb 24	Mobile responsiveness pass; accessibility compliance; analytics disk cache
Feb 25	Atlas feature pack (10 tools); political drill-down; sitemap expansion
Feb 26	Structured field parsing; StarDict dictionary builder; strategic intelligence layers
Feb 27	Encoding corruption repair; v3.0 release
Feb 28	v3.2 release (pipe delimiters, SourceFragment provenance, 18 new parsers); CSI Studies in Intelligence integration; COCOM boundaries
Mar 1	Capital markers; UI polish; atlas performance; source attribution tags
Mar 2	Foreign Leaders dashboards; Development and Inequality analysis; custom Dashboard Builder; rate limiting hardening

1.5 Repository Structure

The project is organized across two repositories:

- **Public archive** (`CIA-World-Factbooks-Archive-1990-2025`): ETL pipeline, raw data, SQLite database, documentation, GitHub Pages landing site. 63 commits.
- **Private webapp** (`cia-factbook-webapp`): FastAPI application, templates, static assets, deployment configuration. 142 commits.

1.6 System Architecture Overview

Figure 1 presents the end-to-end system architecture, from raw CIA source material through the ETL pipeline to the production web application.

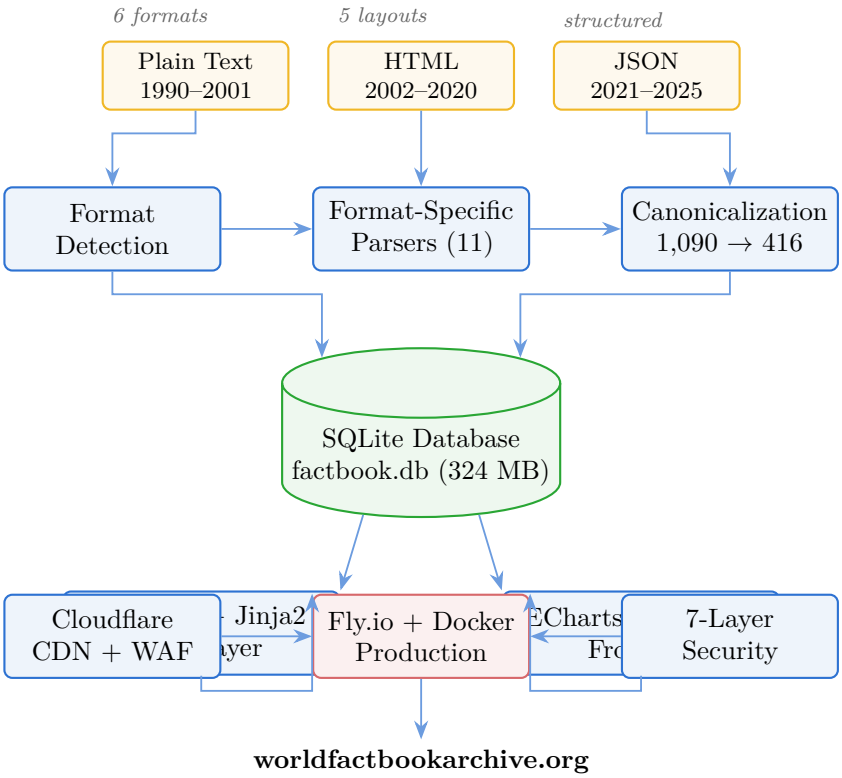

Figure 1.1: System Architecture — from CIA source material to production web application.

Chapter 2

Data Acquisition and Sources

2.1 Source Inventory

The 36 years of Factbook data were obtained from three distinct source channels, each requiring its own ingestion strategy.

2.1.1 Plain Text Editions (1990–2001)

The earliest machine-readable Factbook editions were distributed as plain text files through Project Gutenberg and university FTP mirrors, most notably the University of Missouri archive. These files used six different formatting conventions over the course of a single decade:

Years	Format Name	Structure	Source
1990	`old`	`Country: Name` / `- Section` / `Field: value`	Project Gutenberg
1991	`tagged`	`_@_Name` / `_*_Section` / `_#_Field: value`	University of Missouri
1992	`colon`	`:Country Section` / Field on next line	University of Missouri
1993–1994	`asterisk`	`*Name, Section` / indented `Field:\n value`	University of Missouri
1995–2000	`atsign`	`@Name:Section` / `Field: value`	University of Missouri
2001	`equals`	`@Name` / `Name Section` / `Field: value`	University of Missouri

Each format change reflected internal reorganizations at the CIA's Directorate of Intelligence, which maintained the Factbook.

2.1.2 HTML Editions (2002–2020)

From 2002 onward, the CIA published the Factbook as HTML pages on `cia.gov`. These were retrieved from the Internet Archive's Wayback Machine as ZIP archives, each containing the full set

of country pages for that year.

The HTML structure changed five times over this period:

Years	HTML Pattern	Key Markers
2000	Classic	Named anchors, `<b>Field:</b>`
2001–2008	Table	`<td class="FieldLabel">`, `<a name="SectionID">`
2009–2014	CollapsiblePanel	`<div class="CollapsiblePanel">`, `<span class="category">`
2015–2017	ExpandCollapse	`<h2 class="question" sectiontitle="...">`
2018–2020	Modern	`<div id="field-anchor-*">`, `<div class="category_data subfield">`

2.1.3 JSON Editions (2021–2025)

Beginning in 2021, the CIA published structured JSON data through its website. These were obtained via the `factbook-json-cache` repository, which maintained weekly snapshots of the CIA's JSON feed. Year-specific snapshots were extracted using git history, checking out the last commit before each year's cutoff date.

The final snapshot (2025) was captured before the CIA's February 4, 2026 discontinuation announcement.

2.2 Data Format Evolution

Figure 2 illustrates the timeline of format changes across the 36-year publication history. Each color band represents a distinct parsing strategy required by the ETL pipeline.

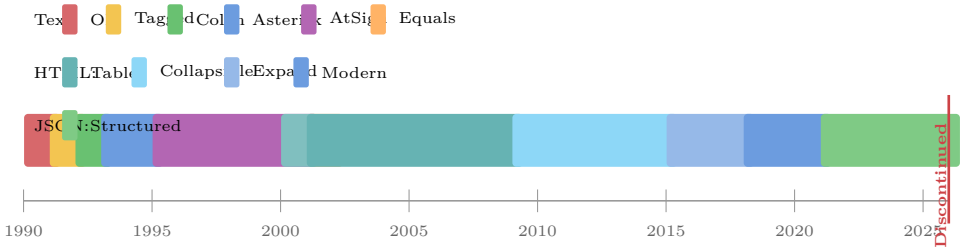

Figure 2.1: Data format evolution across 36 years of CIA World Factbook publication (1990–2025). Each segment represents a distinct parsing strategy.

2.3 Challenges

2.3.1 Encoding Drift

Text files from the 1990s used a mixture of ASCII, Latin-1, and Windows-1252 encodings. Characters such as accented vowels, em-dashes, and special symbols were frequently corrupted during format

conversions. A dedicated repair script (`scripts/repair_encoding_fffd.py`) was developed to detect and correct Unicode replacement characters (`U+FFFD`) in the database.

2.3.2 Entity Name Changes

Over 36 years, dozens of entities changed names, codes, or political status:

- The Soviet Union dissolved into 15 successor states (1991)
- Yugoslavia fragmented into seven countries (1991–2008)
- Swaziland became Eswatini (2018)
- Czechoslovakia split into Czech Republic and Slovakia (1993)
- East Timor became Timor-Leste (2002)
- Burma became Myanmar in CIA usage

The `MasterCountries` table maps all historical name variants to a single canonical identity, enabling cross-year comparisons.

2.3.3 Field Restructuring

The CIA restructured its field taxonomy repeatedly. A field called "National product" in 1990 became "GDP" in the mid-1990s, then "GDP (purchasing power parity)" in the 2000s, and finally "Real GDP (purchasing power parity)" in recent editions. The canonicalization layer (described in Part III) maps all 1,090 field name variants to 416 canonical names.

Chapter 3

ETL Pipeline and Parsing Methodology

3.1 Pipeline Overview

The Extract-Transform-Load pipeline converts raw Factbook source material into a normalized SQLite database through the following stages. Figure 4 illustrates the complete data flow.

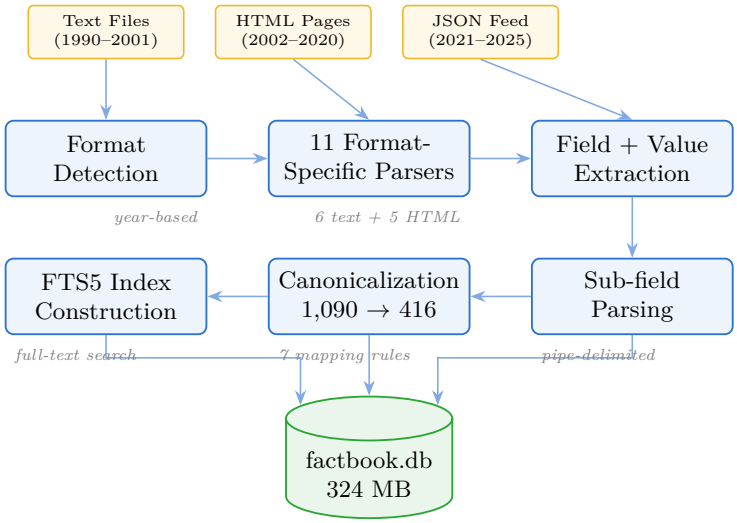

Figure 3.1: ETL pipeline data flow. Raw source material passes through format detection, parsing, sub-field extraction, canonicalization, and indexing before reaching the production SQLite database.

3.2 Text Format Parsers

Six parsers handle the plain text editions from 1990 through 2001. Each parser implements the same interface: accept raw text, return a list of `(country, section, field_name, value)` tuples.

3.2.1 Old Format (1990)

The earliest format uses a simple `Country:` header followed by dash-prefixed section names and inline `Field: value` pairs.

```
Country:  Afghanistan
- Geography
Total area: 647,500 km2
- People
Population: 15,862,293 (July 1990)
```

The parser detects `Country:` lines, splits sections on `- SectionName`, and extracts inline fields using the `extract_inline_fields()` function.

3.2.2 Tagged Format (1991)

A marker-based format using three-character prefixes to denote structural elements:

```
_@_Afghanistan
_*_Geography
_#_Total area: 647,500 km2
_*_People
_#_Population: 16,450,304
```

- `_@_` = Country name
- `_*_` = Section header
- `_#_` = Field entry

Continuation lines (without markers) are appended to the previous field's value.

3.2.3 Colon Format (1992)

Section headers are prefixed with a colon followed by the country name and section, with field values on indented continuation lines:

```
:Afghanistan Geography
Total area:
    647,500 km2
:Afghanistan People
Population:
    16,095,664
```

The parser uses `extract_indented_fields()` to associate indented lines with their parent field names.

3.2.4 Asterisk Format (1993–1994)

This format introduced hierarchical sub-fields with indentation-based nesting:

```
*Afghanistan, Geography
Location:
  South Asia, between Iran and Pakistan
Area:
 total area:
```

```
  647,500 km2
 land area:
  647,500 km2
```

In 1994, the CIA restructured its internal database, causing sub-fields to appear at column 0 instead of indented. The parser distinguishes parent fields from sub-fields using a capitalization convention: parent field names start with a capital letter (`Area`), while sub-field names are lowercase (`total area`, `land area`).

3.2.5 At-Sign Format (1995–2000)

The longest-lived text format, with two sub-variants:

Standard variant (1995, 1997–1998): `@CountryName:SectionName` with inline and indented fields.

```
@Afghanistan:Geography
 Location: Southern Asia, north of Pakistan
 Area:
 total area: 647,500 sq km
```

Bare variant (1996, 1999): `@CountryName` as a standalone line, followed by bare section headers.

The parser uses `extract_mixed_fields()` to handle both inline `Field: value` patterns and indented sub-field structures.

3.2.6 Equals Format (2001)

A transitional format used only for the 2001 edition, where the HTML version was incomplete:

```
@Afghanistan

Afghanistan    Introduction
Background: Afghanistan has been...

Afghanistan    Geography
Location: Southern Asia
```

Country markers appear as `@CountryName`, followed by `CountryName  SectionName` (separated by multiple spaces).

3.3 HTML Parsers

Five HTML parsers handle the web editions from 2000 through 2020, each adapted to the CIA's evolving web design.

3.3.1 Classic HTML (2000)

Uses named anchors (`<a name="Geo">`) for sections and bold tags (`<b>Field:</b>`) for field labels. Values appear as inline text after the closing bold tag.

3.3.2 Table HTML (2001–2008)

Section anchors remain, but fields are organized in HTML tables with `<td class="FieldLabel">` containing field names and sibling `<td>` elements containing values.

3.3.3 CollapsiblePanel HTML (2009–2014)

Sections are wrapped in `<div class="CollapsiblePanel">` containers. Field names appear in `<div class="category">` elements within alternating-row tables. Values appear in `<div class="category_data">` elements.

3.3.4 ExpandCollapse HTML (2015–2017)

Section headers use `<h2 class="question" sectiontitle="Geography">`. Fields are in `<div id="field" class="category">` with `<a>` labels. Values follow in `<div class="category_data">` siblings.

3.3.5 Modern HTML (2018–2020)

The most structured HTML format, using semantic IDs: `<div id="field-anchor-geography-location">` for field labels and `<div id="field-location">` for values. Sub-field data appears in `<div class="category_data subfield text">` elements.

All HTML parsers use the `html_to_pipe_text()` helper function, which converts block-level HTML tags (`<br>`, `</p>`, `</div>`) into pipe separators and strips all remaining markup, producing clean pipe-delimited text suitable for database storage.

3.4 JSON Ingestion (2021–2025)

The JSON editions require minimal parsing, as the CIA provided structured key-value data. The `reload_json_years.py` script checks out year-specific commits from the `factbook-json-cache` repository and extracts country data from the JSON structure into the same `(country, section, field, value)` tuple format used by all other parsers.

3.5 Field Canonicalization

Across 36 years of publication, the CIA used 1,090 distinct field name variants for what are effectively 416 unique concepts. The canonicalization layer (`build_field_mappings.py`) applies seven mapping rules in sequence:

3.5.1 Rule 1: Identity

Fields that appear in the 2024–2025 data unchanged are mapped to themselves.

3.5.2 Rule 2: Dash Normalization

Inconsistent dash formatting (`Economy-overview`, `Economy--overview`) is normalized to `Economy - overview` before applying subsequent rules.

3.5.3 Rule 3: Known Renames

A curated mapping table of 200+ entries handles CIA vocabulary changes:

Original (historical)	Canonical (modern)
Total area	Area
National product	Real GDP (purchasing power parity)
Ethnic divisions	Ethnic groups
Unemployment	Unemployment rate
Military branches	Military and security forces
Comparative area	Area - comparative

3.5.4 Rule 4: Consolidation

Sub-fields that were historically published as separate entries are consolidated under a parent field:

- `Oil - production`, `Oil - consumption`, `Oil - exports`, `Oil - imports` –> **Petroleum**
- `Electricity - production`, `Electricity - consumption` –> **Electricity**

3.5.5 Rule 5: Country-Specific Classification

Detects fields that are specific to individual countries (e.g., government body names from 1990s datasets) and classifies them appropriately.

3.5.6 Rule 6: Noise Detection

Identifies and flags parser artifacts: single-letter field names, lowercase sub-field fragments (`total population`, `male`, `female`), and text fragments mistakenly captured as field names. Approximately 300–400 entries are classified as noise.

3.5.7 Rule 7: Manual Review

Fields not matching any automated rule are flagged for human review.

3.6 Sub-Field Extraction

Multi-line Factbook values are stored as pipe-separated strings, preserving the hierarchical relationship between parent fields and their sub-components:

```
total: 4,250,000 | male: 2,150,000 | female: 2,100,000
agriculture: 28% | industry: 24% | services: 48%
```

This flat representation enables both full-text search and structured queries on sub-field values.

3.7 Encoding Repair

A dedicated repair script scans the database for Unicode replacement characters (`U+FFFD`), which indicate encoding corruption from the original text files. The script cross-references corrupted fields against other years' data for the same country to infer the correct characters, then applies targeted fixes.

Chapter 4

Database Design

4.1 Schema Overview

The archive uses a normalized relational schema in SQLite, organized around five core tables. Figure 3 presents the entity-relationship diagram.

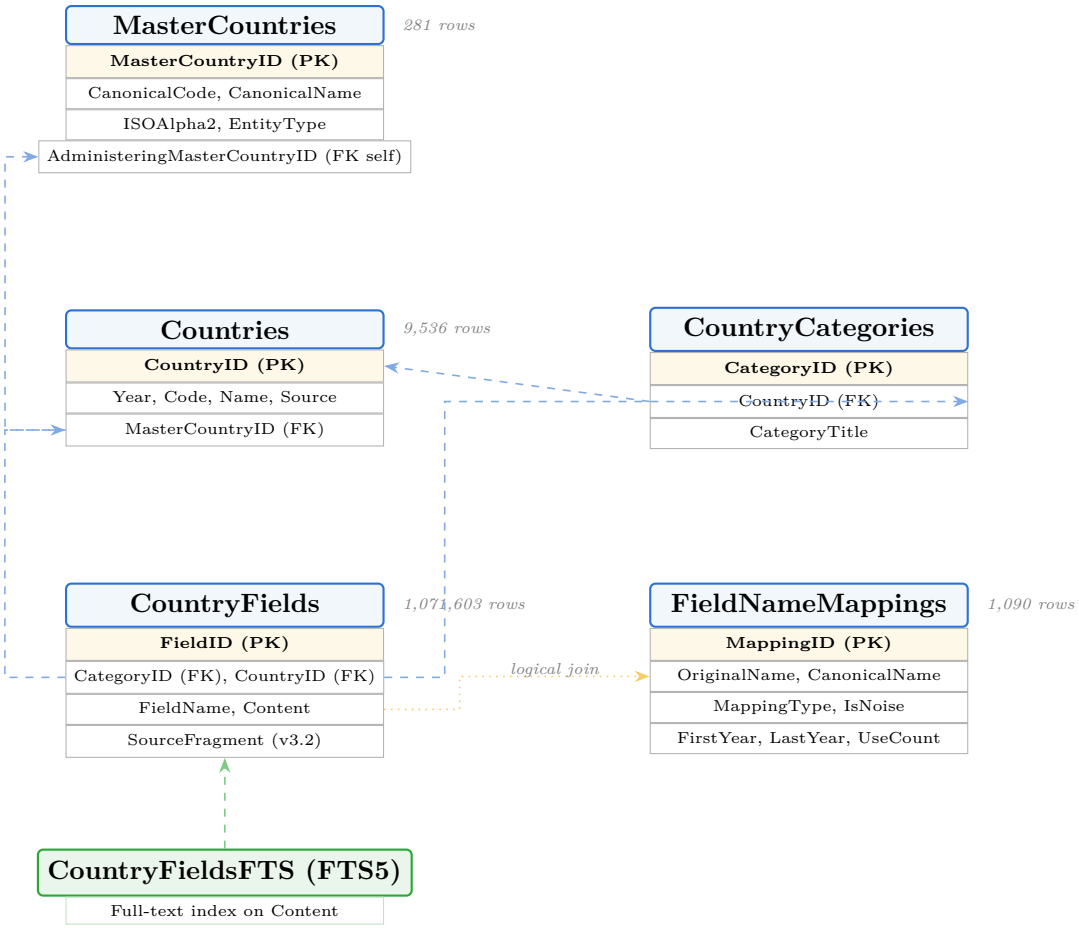

Figure 4.1: Entity-Relationship diagram of the CIA World Factbook Archive database. Dashed arrows indicate foreign key relationships; the dotted gold line shows the logical join between field names and their canonical mappings.

4.2 Design Decisions

4.2.1 Flat Key-Value vs. Wide Table

The archive stores data as key-value pairs (`FieldName` + `Content`) rather than as a wide table with one column per indicator. This design was chosen because:

1. **Schema stability.** The CIA adds, removes, and renames fields across editions. A wide table would require schema migrations for every vocabulary change.
2. **Sparse data.** Not all countries have all fields in all years. A wide table would be predominantly NULL.
3. **Full-text search.** The FTS5 index on `Content` enables search across all field values without knowing the field name in advance.

4.2.2 Pipe-Delimited Sub-Values

Rather than creating a separate sub-field table (which would multiply the row count by 3–5x), sub-field values are stored as pipe-delimited strings within the `Content` column. The webapp's

query layer parses these at read time.

4.2.3 SourceFragment Provenance

Version 3.2 added a `SourceFragment` column to `CountryFields`, preserving the original raw text from the source file before parsing. This enables auditing of parse accuracy and debugging of extraction errors.

4.2.4 SQLite as Production Database

The project migrated from SQL Server to SQLite on February 16, 2026 for several reasons:

- **Portability:** Single-file database deploys inside a Docker container with zero configuration.
- **Performance:** Read-only workload with WAL mode and 512 MB page cache handles concurrent queries efficiently.
- **Cost:** Eliminates the need for a managed database service in production.
- **Size:** The complete database is 324 MB, small enough to bundle in the Docker image.

4.3 Indexes

The following indexes optimize the most common query patterns:

Index	Columns	Purpose
`IX_Countries_Year`	Year	Year-based filtering
`IX_Countries_Code`	Code	Country lookup
`IX_Countries_MasterCountryID`	MasterCountryID	Entity mapping
`IX_Categories_Country`	CountryID	Section retrieval
`IX_Fields_Category`	CategoryID	Field listing
`IX_Fields_FieldName`	FieldName	Field name search
`IX_FieldNameMappings_CanonicalName`	CanonicalName	Canonical name lookup

4.4 Full-Text Search

An FTS5 virtual table (`CountryFieldsFTS`) indexes the `Content` column of all 1,071,603 field records, enabling sub-second full-text search across the entire archive.

4.5 Indicator System

The webapp defines 102 queryable indicators in `fv_indicators.py`, each mapping a human-readable label to a specific `CanonicalName` + `SubField` pair in the database. Indicators are organized into categories:

Category	Count	Examples
Demographics	19	Population, birth rate, life expectancy
Age Structure	6	0–14%, 15–64%, 65+%
Economy	14	GDP (PPP), GDP per capita, inflation, unemployment

Category	Count	Examples
Technology	5	Internet users, mobile subscriptions
Health	8	Infant mortality, physicians density, obesity
Environment	6	CO2 emissions, forest area, renewable energy
Military	5	Military expenditure, active personnel
Governance	4	Corruption index, government spending
Education	3	Literacy, education expenditure
Infrastructure	6	Airports, roadways, railways, ports
Other	26	Area, coastline, elevation, budget

Each indicator carries formatting metadata (number format, units label, sort direction) used by the webapp's charting and table components.

Chapter 5

Web Application Architecture

5.1 Technology Stack

Layer	Technology	Role
Server	FastAPI 0.109+	Async Python web framework
Templates	Jinja2 3.1+	Server-side HTML rendering
Database	SQLite 3	Read-only data store (WAL mode)
Charts	Apache ECharts 5	Interactive data visualization
Maps	Mapbox GL JS	3D globe and choropleth rendering
Styling	Custom CSS (intel-theme)	Dark Intelligence design system
Deployment	Fly.io + Docker	Container hosting with persistent volume
CDN/DNS	Cloudflare	Caching, DDoS protection, Bot Fight Mode

Figure 5 shows the layered architecture of the web application.

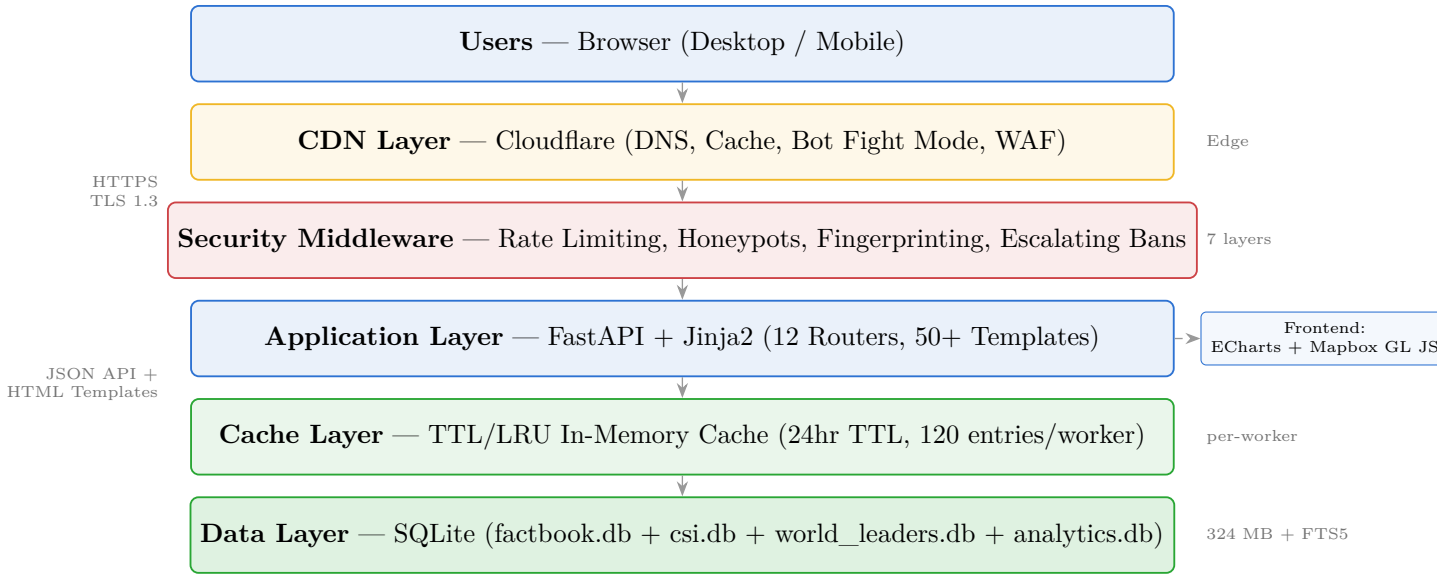

Figure 5.1: Layered architecture of the web application, from user browser through CDN, security middleware, application logic, caching, and data storage.

5.2 Directory Structure

```
cia-factbook-webapp/
  webapp/
    main.py              -- Application entry, security middleware
    database.py          -- SQLite query interface
    cache.py             -- TTL/LRU query cache
    fv_indicators.py     -- 102 indicator definitions
    bot_taxonomy.py      -- User-agent classification
    routers/
      core.py            -- Home, country pages, search
      api.py             -- /api/v2/ JSON endpoints
      analysis.py        -- Analysis pages (rankings, trends, scatter)
      atlas.py           -- Intelligence Atlas
      csi.py             -- CIA Studies in Intelligence
      political.py       -- Political regime analysis
      world_leaders.py   -- Foreign leaders dashboards
      development.py     -- Development and Inequality
      export.py          -- CSV/Excel bulk export
    templates/           -- 50+ Jinja2 templates
    static/
      css/intel-theme.css
      js/dg2-echarts.js
  start.py               -- Docker startup (DB sync + Uvicorn)
  fly.toml               -- Fly.io deployment manifest
  Dockerfile             -- Container build
```

```
deploy.sh              -- Safe deploy script
```

5.3 Routing Architecture

The application is organized into 12+ routers, each handling a specific domain:

Router	Prefix	Responsibility
`core.py`	`/`, `/country/`	Home page, country archives, global search
`api.py`	`/api/v2/`	JSON API for rankings, timeseries, scatter, compare
`analysis.py`	`/analysis/`	Rankings, trends, scatter, changes, compare, query builder, dashboard builder
`atlas.py`	`/analysis/atlas`	Full-screen 3D globe with geopolitical overlays
`csi.py`	`/analysis/csi/`	CIA Studies in Intelligence reading room
`political.py`	`/analysis/political`	Political regime dashboard
`world_leaders.py`	`/analysis/foreign-leaders`	Foreign leadership analysis
`development.py`	`/analysis/development`	Development and inequality metrics
`export.py`	`/export/`	CSV and Excel bulk data export

5.4 API Layer

The `/api/v2/` endpoints provide JSON data to the frontend charts and tables:

Endpoint	Method	Description
`/api/v2/rank/{indicator}/{year}`	GET	Ranked list of countries by indicator value
`/api/v2/timeseries/{indicator}/{code}`	GET	Time series for one country
`/api/v2/compare/{year}`	GET	Side-by-side comparison of selected countries
`/api/v2/scatter`	GET	X-Y scatter data for two indicators
`/api/v2/search`	GET	Full-text search across all fields
`/api/v2/changes/{indicator}`	GET	Year-over-year changes
`/api/v2/countries`	GET	Country list with metadata

5.5 Caching Strategy

A TTL/LRU cache (`webapp/cache.py`) sits between the routers and the database. Since the archive data is read-only historical data that only changes on deployment, aggressive caching is safe:

- **TTL:** 24-hour time-to-live on all cached queries
- **LRU:** Maximum 120 entries per worker; oldest evicted on overflow
- **Scope:** Each Uvicorn worker process maintains its own independent cache
- **Effect:** Reduces 5–30 second analysis queries to under 0.5 seconds on cache hit
- **Invalidation:** Manual via `/admin/clear-cache?key=ADMIN_KEY`

Chapter 6

Design System: Dark Intelligence

6.1 Philosophy

The visual design draws inspiration from intelligence operations centers and classified document aesthetics. The "Dark Intelligence" theme uses a dark blue-grey palette with high-contrast text and cyan accent colors, conveying the seriousness and precision appropriate to intelligence data.

6.2 Color Palette

6.2.1 Backgrounds and Surfaces

Token	Hex	Usage
`--bg-body`	`#111418`	Page background
`--bg-surface`	`#1C2127`	Card and panel backgrounds
`--bg-elevated`	`#252A31`	Elevated surfaces (modals, dropdowns)
`--border`	`#2F343C`	Standard borders
`--border-strong`	`#383E47`	Emphasized borders

6.2.2 Text Hierarchy

Token	Hex	Usage
`--text-bright`	`#FFFFFF`	Maximum emphasis (stat values)
`--text-heading`	`#E0E4E8`	Section headings
`--text-primary`	`#C5CBD3`	Primary body text
`--text-body`	`#ABB3BF`	Standard body text
`--text-muted`	`#7E8A98`	De-emphasized text

6.2.3 Accent Colors

Token	Hex	Usage
`--accent-blue`	`#2D72D2`	Primary actions, links
`--accent-gold`	`#F0B726`	Highlights, search results
`--accent-red`	`#CD4246`	Errors, critical values
`--accent-green`	`#29A634`	Success, positive changes

6.3 Component Classes

The design system provides reusable CSS classes for all UI components, ensuring visual consistency across 50+ templates:

- **`.stat-card`**: Dashboard statistics with background surface color and subtle shadow
- **`.data-table`**: Tabular data with alternating row backgrounds
- **`.filter-bar`**: Horizontal filter toolbars with inline controls
- **`.badge-*`**: Status badges (moderate, low, critical, NA)
- **`.btn`**: Button variants with hover and active states
- **`.form-control`**, **`.form-label`**: Form inputs styled for dark backgrounds
- **`.diff-*`**: Change highlighting (added, removed, modified)

6.4 Typography

- **Font family:** Georgia, serif — chosen for its formal, document-like quality
- **Base size:** 1.05rem with 1.65 line height
- **Rendering:** Antialiased with `text-rendering: optimizeLegibility`

6.5 Motion

All interactive transitions use a consistent 200ms duration with cubic-bezier easing, providing responsive feedback without distraction.

6.6 Responsive Design

The layout adapts to three breakpoints:

- **Desktop** (>1200px): Full three-panel layouts, side-by-side charts
- **Tablet** (768–1200px): Stacked panels, reduced chart widths
- **Mobile** (<768px): Single-column layout, collapsible controls, hamburger navigation

6.7 Dual-Theme Support

While the dark theme is the default and recommended experience, a light theme override is available via `[data-theme="light"]`. Certain elements (topbar, charts, maps) remain dark in both themes for visual consistency.

Chapter 7

Feature Guide

This section documents each feature of the web application, including its purpose, how to use it, and what data it presents.

7.1 Home Page

URL: `/`

The landing page presents a global search bar, summary statistics (281 entities, 36 editions, 9,536 records, 1,071,603 data points), and navigation cards linking to the major sections of the application.

Usage: Type any country name, indicator, or keyword into the search bar to find relevant data across all years. Click navigation cards to access specific analysis tools.

7.2 Country Pages

URL: `/country/{code}/{year}`

Each country has a dedicated page showing the complete Factbook entry for that year, organized by section (Geography, People, Economy, Government, Military, etc.).

Usage:

1. Select a country from the dropdown or search.
2. Use the year navigation bar to move between editions.
3. Click section headers to expand or collapse field groups.
4. Each field shows its value as published in that year's Factbook.

Data shown: All fields from the selected country-year record, organized by CIA section categories.

7.3 Country Compare

URL: `/analysis/compare`

Side-by-side comparison of two or more countries on key indicators for a selected year.

Usage:

1. Select 2–10 countries using the country picker (type to search, click to add).

2. Choose a year from the dropdown.
3. The comparison table shows all available indicators for the selected countries, with values aligned in columns.

Data shown: 102 queryable indicators with formatted values, units, and color-coded ranking.

7.4 Rankings

URL: `/analysis/rankings`

Horizontal bar chart ranking all countries by a selected indicator for a given year.

Usage:

1. Choose an indicator from the categorized dropdown (Demographics, Economy, Military, etc.).
2. Select a year.
3. Optionally filter by COCOM region (AFRICOM, CENTCOM, EUCOM, INDOPACOM, SOUTHCOM, NORTHCOM) or adjust the Top N slider (5–50).
4. The bar chart updates to show ranked countries with their values.

Data shown: Ranked values for the selected indicator, with country flags and formatted statistics.

7.5 Trends

URL: `/analysis/trends`

Multi-country time series line chart showing how an indicator changes over the 36-year archive span.

Usage:

1. Select an indicator.
2. Add 1–10 countries to compare.
3. Adjust the year range slider to focus on a specific period.
4. Hover over data points for exact values; scroll to zoom.

Data shown: Year-by-year values for each selected country, plotted as line series with distinct colors.

7.6 Scatter Analysis

URL: `/analysis/scatter`

X-Y scatter plot correlating two indicators across all countries for a selected year.

Usage:

1. Choose an X-axis indicator (e.g., GDP per capita).
2. Choose a Y-axis indicator (e.g., Life expectancy).
3. Select a year.
4. Each bubble represents a country; bubble size reflects population.
5. Click a bubble to identify the country; hover for details.
6. A regression line shows the overall trend.

Data shown: Two-dimensional correlation of indicators across all sovereign nations, with statistical regression.

7.7 Changes

URL: `/analysis/changes`

Year-over-year change analysis showing which countries experienced the largest absolute or percentage changes in a selected indicator.

Usage:

1. Choose an indicator.
2. Select two years to compare (e.g., 2020 vs. 2025).
3. Toggle between absolute change and percentage change.
4. The bar chart highlights the biggest movers in both directions.

Data shown: Sorted list of countries by magnitude of change, with before/after values.

7.8 Query Builder

URL: `/analysis/query-builder`

A SQL-like visual interface for constructing custom indicator queries with filters, sorting, and export.

Usage:

1. Select one or more indicators from the categorized dropdown.
2. Add country filters (specific countries, regions, or entity types).
3. Set year range and sorting preferences.
4. Click "Run Query" to execute.
5. Results appear in a sortable, downloadable table.

Data shown: Custom-filtered indicator data with export to CSV.

7.9 Dashboard Builder

URL: `/analysis/dashboard-builder`

A drag-and-drop canvas for composing personalized analytical dashboards from 10 widget types.

Usage:

1. **Add widgets:** Click widget types in the left palette (KPI Card, Rankings Bar, Time Series, Scatter Plot, Compare Table, Data Table, Country Profile, Field Text, Mini Globe, Field History).
2. **Configure:** Click the gear icon on any widget to open the config panel. Select indicators, countries, years, and other parameters. Click "Apply."
3. **Layout:** Drag widgets to reorder. Use the resize button to toggle between column spans (25%, 33%, 50%, 67%, 100%). Use up/down arrows to move widgets.
4. **Presets:** Use the toolbar dropdowns to load pre-built dashboards (Deep Dive, Factbook Layout) for any country and year.
5. **Persistence:** Dashboards auto-save to localStorage. Use Export JSON / Import JSON to share layouts.
6. **Apply All:** The "Apply Country to All" and "Apply Year to All" buttons push a single country or year setting to every compatible widget.

Widget types:

Widget	Description	Data Source
KPI Card	Single stat with label	`/api/v2/rank/{ind}/{year}?limit=1`
Rankings Bar	Horizontal bar chart	`/api/v2/rank/{ind}/{year}?limit=N`
Time Series	Multi-country line chart	`/api/v2/timeseries/{ind}/{code}`
Scatter Plot	X-Y correlation	`/api/analysis/scatter`
Compare Table	Multi-country comparison	`/api/v2/compare/{year}`
Data Table	Full ranked table	`/api/v2/rank/{ind}/{year}`
Country Profile	Key stats summary	Multiple API calls
Field Text	Raw factbook text	`/api/v2/field/{code}/{year}`
Mini Globe	Interactive 3D globe	Mapbox GL JS + GeoJSON
Field History	Year-over-year text diff	Multiple year API calls

7.10 Intelligence Atlas

URL: `/analysis/atlas`

A full-screen interactive 3D globe with geopolitical data overlays, measurement tools, and strategic intelligence layers.

Usage:

1. **Navigate:** Click and drag to rotate the globe; scroll to zoom; right-click drag to tilt.
2. **Choropleth:** Select an indicator from the sidebar to color countries by value.
3. **Click:** Click any country to view its data card with key statistics.
4. **Layers:** Toggle strategic layers: COCOM boundaries, missile sites, military facilities, capital markers, state/province borders.
5. **Tools:** Use the measurement tool for distance/area calculations, the screenshot tool for image export, and the range ring tool for radius analysis.
6. **Projection:** Toggle between globe and flat map projections.

Data shown: 102 indicators as choropleth overlays, international disputes, infrastructure data (airports, ports), missile capabilities (OSINT), capital coordinates, COCOM regional boundaries.

7.11 CSI Reading Room

URL: `/analysis/csi/dashboard`

A searchable digital library of CIA *Studies in Intelligence* journal articles, with full-text search, entity analysis, and collection management.

Usage:

1. **Search:** Enter keywords in the search bar to find articles. Filters available for year range and category.
2. **Browse:** Navigate by year/issue or category.
3. **Read:** Click any article to view its full text in a cleaned reader view.
4. **Collections:** Save articles to local collections (stored in localStorage). Export collections for offline reference.

5. **Dashboard:** View analytics on publication trends, category distribution, and keyword frequency over time.

Data shown: CIA Studies in Intelligence articles with metadata (author, date, category, issue), full-text content, entity co-occurrence graphs, and publication analytics.

7.12 StarDict Dictionary

An offline companion product: the complete Factbook archive packaged as a StarDict-format dictionary compatible with GoldenDict (desktop) and KOReader (e-readers).

Usage: Download the StarDict files from the GitHub repository and import into any compatible dictionary application. Look up any country name to see its full Factbook entry.

7.13 Search

URL: `/` (search bar) or `/search?q=...`

Global full-text search powered by SQLite FTS5, spanning all 1,071,603 field records across all years and entities.

Usage: Type any term into the search bar. Results are grouped by country and year, with matching text highlighted. Click any result to navigate to the full country page.

Chapter 8

Infrastructure and Deployment

8.1 Hosting Architecture

The application runs on **Fly.io**, a container hosting platform, in the `iad` (US East Coast) region.

Component	Configuration
Machine	Single Fly.io machine, auto-start/stop
Workers	2 Uvicorn worker processes
Port	8080 (internal)
Volume	1 GB persistent volume at `/data`
Health check	`/health` endpoint, 30s interval, 15s timeout

8.2 Docker Container

The application is packaged as a Docker container based on `python:3.12-slim`:

1. Install Python dependencies from `requirements.txt`
2. Copy application code and static assets
3. Bundle SQLite databases in the image (factbook.db, csi_studies_index.sqlite, world_leaders_structured.sqlite)
4. Bundle IP2Location geolocation binary
5. Entry point: `python start.py`

The `start.py` startup script performs MD5 hash comparison between bundled databases and the persistent volume, copying updated databases only when the hash changes. This ensures database updates propagate on deployment without unnecessary I/O.

8.3 Deployment Workflow

The `deploy.sh` script implements a git-aware safe deploy:

1. Check for uncommitted changes (`git diff`, `git diff --cached`)
2. If dirty: stash changes with timestamp (`deploy-safeguard-{ts}`)
3. Run `fly deploy` (builds from local filesystem)
4. Pop stash to restore local work

This prevents accidental deployment of unfinished code while preserving work-in-progress.

8.4 DNS and CDN

Cloudflare provides DNS resolution, HTTP proxy, and security for `worldfactbookarchive.org`:

- DNS proxied through Cloudflare (orange cloud)
- Bot Fight Mode enabled (blocks known bad bots at the edge)
- SSL/TLS: Full (strict) mode
- Cache: Static assets cached at Cloudflare edge; HTML served fresh

The legacy domain `cia-factbook-archive.fly.dev` returns 301 redirects to the custom domain.

8.5 Security Architecture

The application implements a 7-layer security system:

8.5.1 Layer 1: Honeypot Endpoints

Paths like `/admin`, `/wp-admin`, `/.env`, and `/database` are disallowed in `robots.txt`. Legitimate crawlers obey robots.txt; only malicious bots follow these paths. Accessing a honeypot triggers an instant IP ban.

8.5.2 Layer 2: User-Agent Blocking

Known bot user-agent strings (python-requests, scrapy, wget, httpx) are blocked on sight. AI training crawlers (Claude, GPT, Grok, Perplexity) are blocked entirely.

8.5.3 Layer 3: Request Fingerprinting

The system analyzes request timing variance per IP. Humans produce irregular intervals (3.2s, 47.1s, 2.8s); bots produce uniform intervals (<1s variance). IPs with 8+ requests showing bot-like timing patterns are auto-banned.

8.5.4 Layer 4: Per-Endpoint Rate Limits

Targeted limits per path prefix:

Path	Limit	Window
`/export/`	10 requests	60 seconds
`/analysis/`	20 requests	60 seconds
`/archive/`	20 requests	60 seconds
`/api/`	60 requests	60 seconds

Exceeding limits returns HTTP 429 and records a strike.

8.5.5 Layer 5: Escalating Bans

After 15 strikes within 5 minutes, the IP is banned. Ban duration escalates on repeat offenses: 5 minutes, 30 minutes, 2 hours, 24 hours.

8.5.6 Layer 6: Ban Persistence

Ban data is saved to `/data/bandata.json` on the persistent volume and reloads on server restart. Restarting the application alone does not clear bans.

8.5.7 Layer 7: Server-Wide Throttle

A global rate limit of 120 page requests per minute across all IPs prevents distributed flooding. If exceeded, the server returns HTTP 503 (Service Busy) to shed load.

8.5.8 Admin Bypass

Authenticated administrators bypass all security layers via an `ADMIN_KEY` stored as a Fly.io secret. Authentication methods: HTTP header, browser cookie (1-year expiry), or query parameter.

Admin endpoints:

Endpoint	Purpose
`/admin/unlock`	Set admin cookie
`/admin/clear-bans`	Clear all ban data
`/admin/clear-cache`	Clear query cache
`/admin/status`	View active bans and cache stats

Chapter 9

Supplementary Materials

9.1 API Reference

All API endpoints accept GET requests and return JSON responses.

Endpoint	Parameters	Response
GET /api/v2/rank/{indicator}/{year}	limit, region, desc	[{rank, name, iso2, value, units}]
GET /api/v2/timeseries/{indicator}/{code}	start_year, end_year	[{year, value, units}]
GET /api/v2/compare/{year}	codes (comma-separated ISO2)	[{name, iso2, indicators: {...}}]
GET /api/v2/scatter	x, y, year	[{name, iso2, x_value, y_value, size}]
GET /api/v2/search	q, year, limit	[{country, year, field, snippet}]
GET /api/v2/changes/{indicator}	year_a, year_b	[{name, iso2, old, new, change, pct}]
GET /api/v2/countries	(none)	[{name, iso2, code, entity_type}]
GET /api/atlas/disputes/{year}	(none)	[{iso2, name, text}]
GET /api/atlas/infrastructure/{year}	(none)	[{iso2, airports, ports, forces, ...}]
GET /api/atlas/missiles	(none)	[{name, iso2, tier, systems, warheads, range}]
GET /api/csi/search	q, year, category, page	[{doc_id, title, date, snippet}]
GET /api/csi/dashboard	year_start, year_end	{doc_count, categories, trends}

9.2 Indicator Categories

The complete list of 102 queryable indicators organized by category:

Demographics (19): Population, Population growth rate, Birth rate, Death rate, Net migration rate, Infant mortality rate (total/male/female), Life expectancy at birth (total/male/female), Total fertility rate, Median age, Urban population percentage, Sex ratio, Dependency ratio (youth/elderly/total)

Age Structure (6): 0–14 years, 15–24 years, 25–54 years, 55–64 years, 15–64 years, 65 years and over

Economy (14): Real GDP (PPP), GDP per capita (PPP), GDP (official exchange rate), Real GDP growth rate, Inflation rate, Unemployment rate, Youth unemployment rate, Public debt, Budget surplus/deficit, Current account balance, Exports, Imports, Gini index, Population below poverty line

Technology (5): Internet users, Broadband subscriptions, Mobile cellular subscriptions, Telephones (fixed lines), Secure internet servers

Health (8): Infant mortality rate, Physicians density, Hospital bed density, Obesity rate, HIV/AIDS prevalence, Drinking water access, Sanitation access, Health expenditure

Environment (6): CO2 emissions, Forest area, Agricultural land, Renewable energy, Electricity access, Revenue from natural resources

Military (5): Military expenditures (% GDP), Active military personnel, Military service age, Military branches count, Arms procurement

Governance (4): Corruption perceptions, Government spending, Political stability, Regulatory quality

Education (3): Literacy rate, Education expenditures, School life expectancy

Infrastructure (6): Airports, Roadways, Railways, Waterways, Ports, Merchant marine

Other (26): Area, Coastline, Elevation (highest/lowest), Climate zones, Natural hazards, Natural resources, Land use, Budget revenues/expenditures, Taxes, Fiscal year, Exchange rates, and more

Chapter 10

Credits and Acknowledgments

10.1 Data Sources

- **Central Intelligence Agency.** *The World Factbook.* Washington, DC: CIA, 1990–2025. Public domain.
- **University of Missouri.** Historical Factbook text file archive (1990s editions).
- **Internet Archive / Wayback Machine.** Historical CIA.gov HTML snapshots (2000–2020).
- **factbook-json-cache.** GitHub repository providing structured JSON snapshots (2021–2025).
- **IP2Location.** LITE geolocation database for visitor analytics.
- **Natural Earth.** GeoJSON boundary data for atlas visualization.

10.2 Open-Source Libraries

Library	Role
FastAPI	Web framework
Jinja2	Template engine
Uvicorn	ASGI server
SQLite / FTS5	Database and full-text search
Apache ECharts	Interactive charting
Mapbox GL JS	3D globe and map rendering
Beautiful Soup	HTML parsing
Alpine.js	Lightweight frontend interactivity
openpyxl	Excel export generation

10.3 Contributors

- **Milan Milkovich** — Project creator, ETL pipeline development, web application development, infrastructure deployment, and design.
- **Claude (Anthropic)** — AI development partner for ETL pipeline engineering, web application features, security architecture, database optimization, documentation, and code review across 200+ development sessions.
- Community members who reported issues and suggested improvements via GitHub Issues.

10.4 Contact

- **Website:** https://worldfactbookarchive.org
- **GitHub:** https://github.com/MilkMp/CIA-World-Factbooks-Archive-1990-2025

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